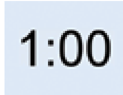
Air Force One
Here's a detailed picture depicting the design and function of the Air Force One.
http://dgit.in/AF1works



One Minute Lists
One minute list compiles a list everyday with interesting little facts and information.
http://dgit.in/oneminlist

GIGABYTE X299 AORUS Gaming 9 VS ASUS PRIME X299-DELUXE

Two of the most feature rich X299 motherboards go head to head.

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Intel's latest and greatest i.e. the X299 HEDT platform arrived a little earlier than scheduled and it's brought with it quite a few powerful processors. With AMD's Threadripper platform serving as competition, Intel seems to have hurriedly pushed out the X299 platform to meet Threadripper head on. Both platforms have excellent processors and board manufacturers have been busy coming out with equally good motherboards to make the most out of them. While Threadripper is still fresh in the market, we do have X299 motherboards pouring in, out of which we have two from the top end. From GIGABYTE we have the X299 AORUS Gaming 9 which is part of their gaming focused lineup and we have the PRIME X299-DELUXE from ASUS.

As flagship boards, both of these support Intel's HEDT CPUs with the maximum PCIe lane count. So we're going to see a slight variation in the choice of peripherals and components that utilise these lanes but in either case, we're going to have all 44 lanes being utilised. Let's see how these boards fare against each other.

BUILD

GIGABYTE X299 AORUS GAMING 9

We start off with one of the most important aspects of a high-end motherboard, the VRM. We see that the Gaming 9 has 8-phases for the CPU VCore which can be seen at the very top of the motherboard. They're using an Infineon IR35201 as the digital PWM with IR3556 MOSFET rated at 50A for each of the individual phases. The inductors on each phase also happen to be rated for 70-76A which should easily handle peak loads of the upcoming 18-core CPU without overclocking. For the RAM we see two IR3553 and two IR35204. Overall, a pretty well-knit design drawing power from two 8-pin ATX connectors.

ASUS PRIME X299-DELUXE

Just as we saw with the Gaming 9, the ASUS PRIME X299-DELUXE uses an Infineon IR35201 as a digital PWM with 8-phases but each phase is being controlled by an IR3555 MOSFET which are rated for 60A which is greater than what the GIGABYTE motherboard has. As for the RAM, we see two Digi+ ASP1103 controllers which is only found on ASUS boards and since there are no publicly available datasheets, we can't get into the nitty gritty of it. Another thing to note is that this VRM circuitry draws power from one 8-pin and one 4-pin ATX power connector which under peak loads will exhibit higher temperatures compared to the GIGABYTE board.